

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

1. Sketch the graph of e^x .
2. Sketch the graph of $\ln(x)$.
3. Explain the relation between the function $f(x) = e^x$ and the function $g(x) = \ln(x)$.
4. Give the domain of $\log_b(x)$, where $b > 0$ and $b \neq 1$.
5. Give the range of $\log_b(x)$, where $b > 0$ and $b \neq 1$.
6. Simplify the expression

$$\frac{(e^{-5x}e^{x^2}\sqrt{e})^2}{e^4}.$$

Exercise 1.

Recall that the equality $y = \log_b(x)$ is equivalent to the equality $x = b^y$. Recall the equalities $\log_b(b^x) = x$ and $b^{\log_b(x)} = x$. Use these to solve the following equations.

1. $2^{3x} = 5$.
2. $\ln(x+1) = 3$.

Exercise 2.

Solve the following equations.

1. $\log_{10}(x-1) + \log_{10}(5) = \log_{10}(20)$.
2. $e^{2\ln(x)} = 7$.
3. $\ln(3e^{2x}) = 4$.